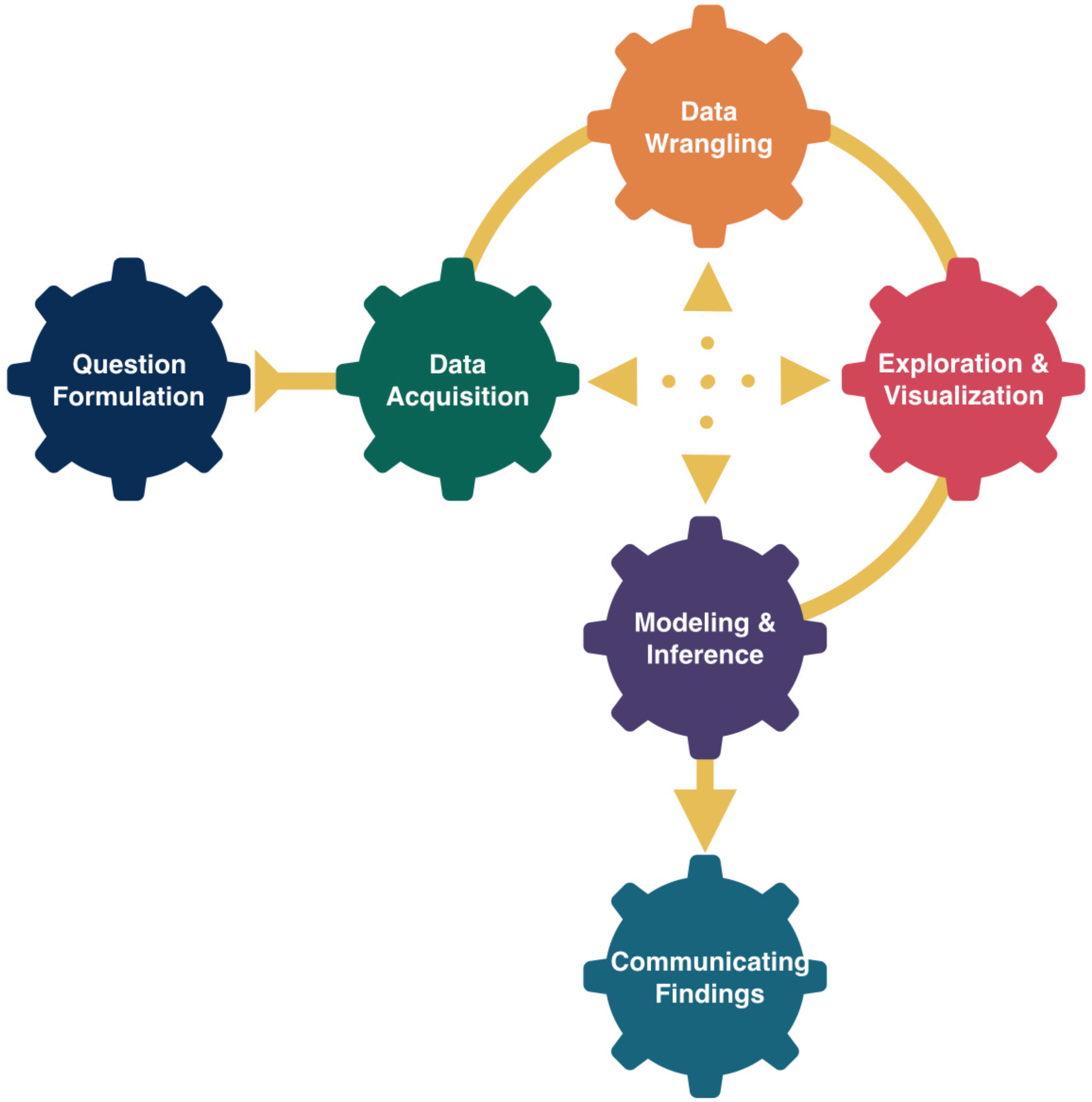

More Theory-Based Inference

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DATA 100

Week 11 | Spring 2026

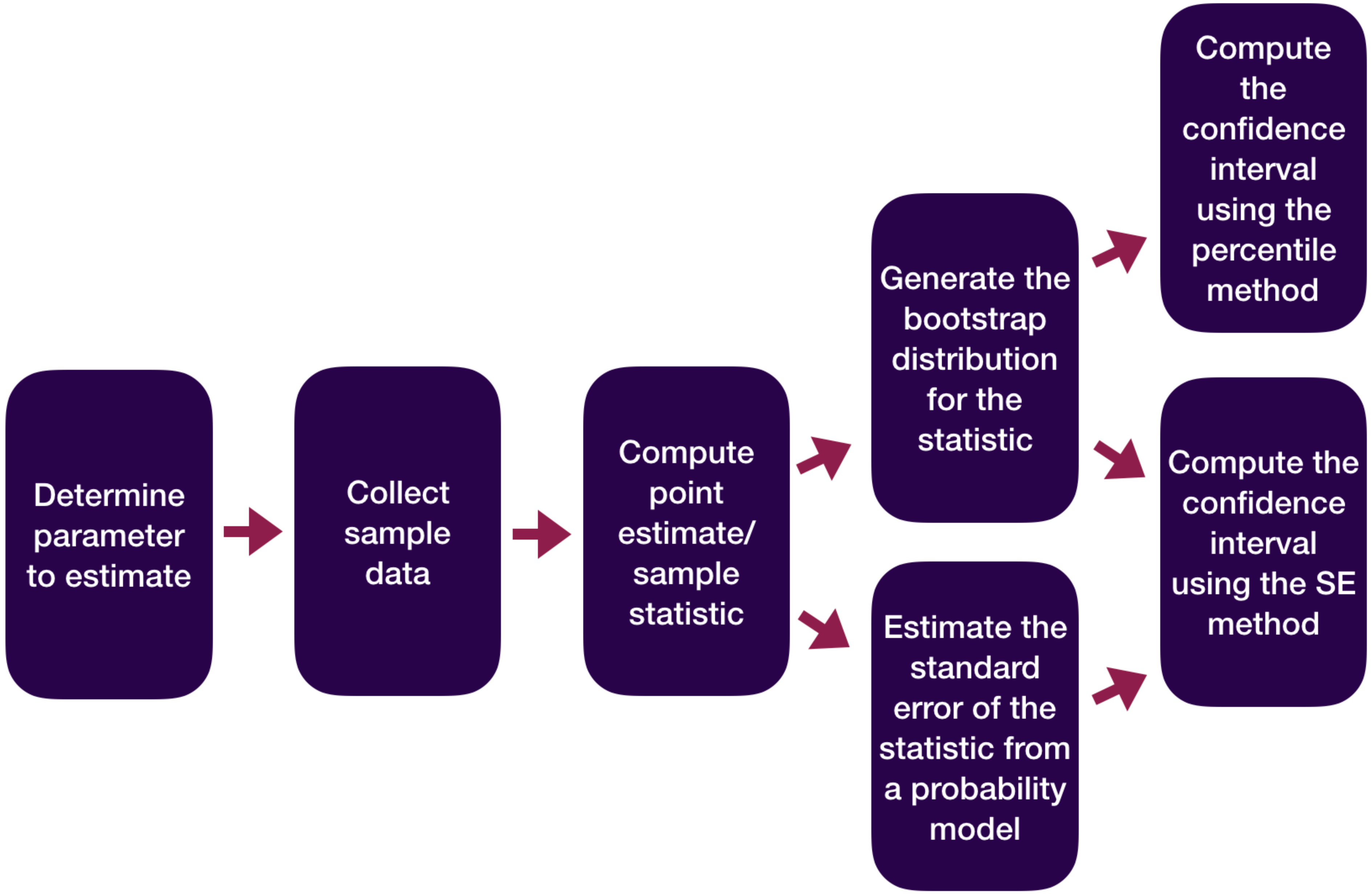
Announcements

- Exam this week:
  - In-class on Wed
  - Oral on Fri. (Sign-ups are in Slack.)

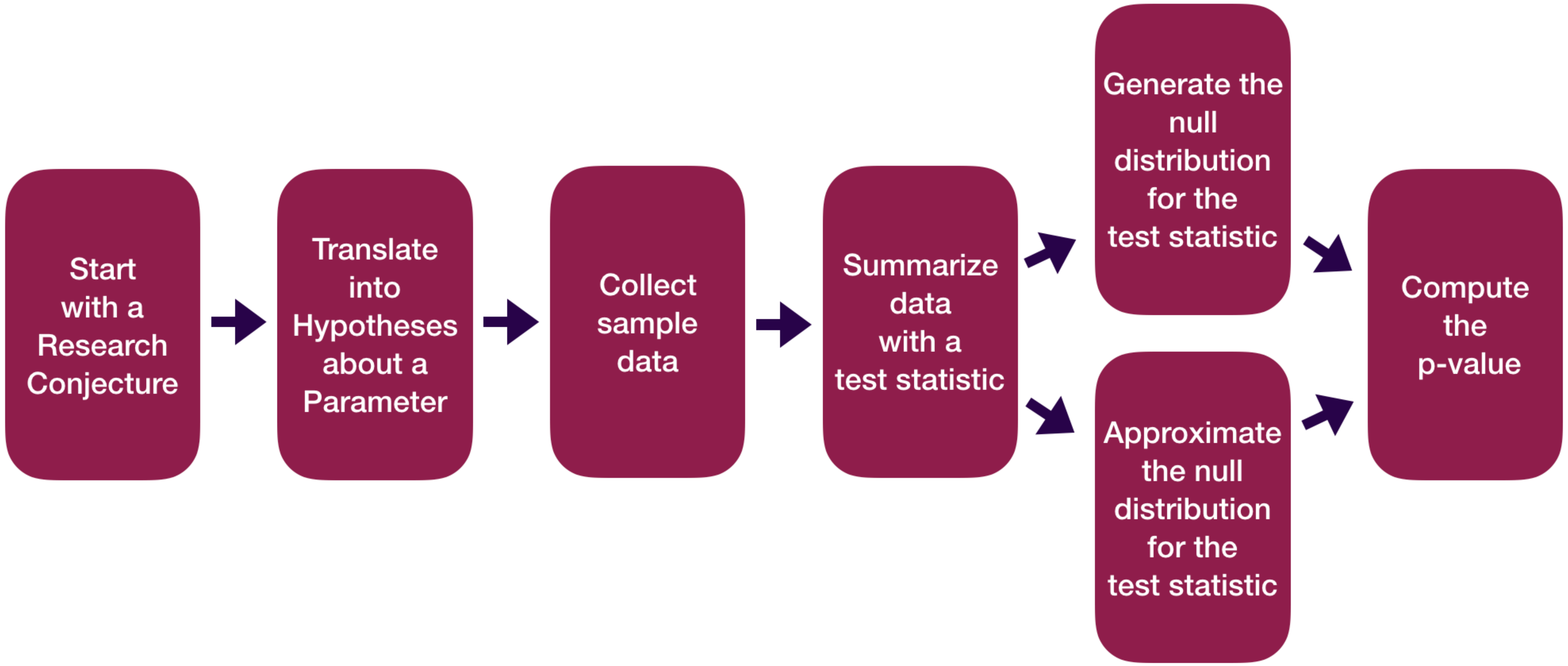
Goals for Today

- Discuss more random variable-based inference.
- Compare methods of inference.

Statistical Inference Zoom Out – Estimation



Statistical Inference Zoom Out – Testing



Recap:

Central Limit Theorem (CLT): For random samples and a large sample size (n), the sampling distribution of many sample statistics is approximately normal.

- Applies to most of the statistics we have explored so far:
  - Sample mean
  - Sample proportion
  - Sample difference in means
  - Sample difference in proportions
  - Correlation

There Are Several Versions of the CLT!

Response	Explanatory	Numerical_Quantity	Parameter	Statistic
quantitative	-	mean	μ	$\bar{x}$
categorical	-	proportion	p	$\hat{p}$
quantitative	categorical	difference in means	$\mu_1 - \mu_2$	$\bar{x}_1 - \bar{x}_2$
categorical	categorical	difference in proportions	$p_1 - p_2$	$\hat{p}_1 - \hat{p}_2$
quantitative	quantitative	correlation	ρ	r

- Refer to these tables for:
    - CLT's "large sample" assumption
    - If you want to know the theory-based R function.
    - If you are curious about the equations for the test statistic and confidence interval

Let’s cover examples of theory-based inference for two variables.

Data Example

We have data on a random sub-sample of the 2010 American Community Survey. The American Community Survey is given every year to a random sample of US residents.

```r
# Libraries
library(tidyverse)
library(Lock5Data)

# Data
data(ACS)
# Focus on adults
ACS_adults <- filter(ACS, Age >= 18)

glimpse(ACS_adults)
```

```
Rows: 1,936
Columns: 9
$ Sex             <int> 0, 1, 0, 0, 1, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 1, …
$ Age             <int> 38, 18, 21, 55, 51, 28, 46, 80, 62, 41, 37, 42, 69, 48…
$ Married         <int> 1, 0, 0, 1, 0, 0, 0, 0, 1, 1, 0, 0, 0, 1, 1, 1, 0, 0, …
$ Income          <dbl> 64.0, 0.0, 4.0, 34.0, 30.0, 13.7, 114.0, 0.0, 0.0, 0.0…
$ HoursWk         <int> 40, 0, 20, 40, 40, 40, 60, 0, 0, 0, 40, 42, 0, 60, 0, …
$ Race            <fct> white, black, white, other, black, white, white, white…
$ USCitizen       <int> 1, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, …
$ HealthInsurance <int> 1, 1, 1, 0, 1, 0, 1, 1, 1, 1, 0, 1, 1, 1, 1, 1, 1, 0, …
$ Language        <int> 1, 1, 1, 0, 1, 0, 0, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, …
```

Difference in Proportions

Let's try to determine if there's a relationship between US citizenship and marriage status.

Response variable:

Explanatory variable:

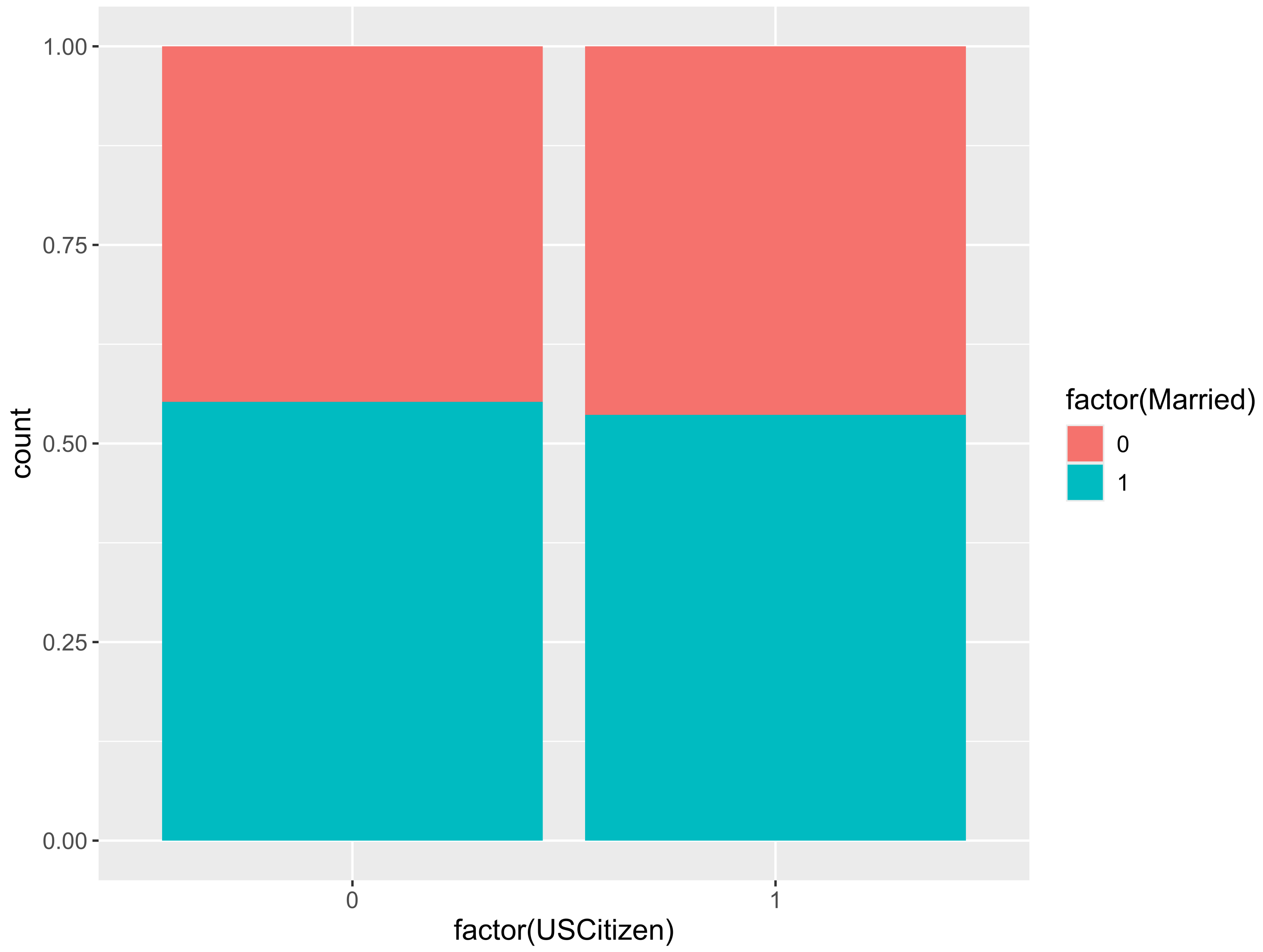
Parameter of interest:

Sample size requirement for theory-based inference:

Difference in Proportions

Let's try to determine if there's a relationship between US citizenship and marriage status.

```r
# Exploratory data analysis
ggplot(data = ACS_adults,
       mapping = aes(x = factor(USCitizen),
                     fill  = factor(Married))) +
  geom_bar(position = "fill")
```




```r
# Sample size
ACS_adults %>%
  count(Married, USCitizen)
```

	Married	USCitizen	n
1	0	0	64
2	0	1	832
3	1	0	79
4	1	1	961

Difference in Proportions

Let's try to determine if there's a relationship between US citizenship and marriage status.

Why is `prop_test()` failing?

```
library(infer)
ACS_adults %>%
prop_test(Married ~ USCitizen,
          order = c("1", "0"), z = TRUE,
          success = "1")
```

```
Error in `prop_test()`:
! The response variable of `Married` is not appropriate since the
  response variable is expected to be categorical.
```

Difference in Proportions

Let's try to determine if there's a relationship between US citizenship and marriage status.

```
ACS_adults %>%
  mutate(MarriedCat = case_when(Married == 0 ~ "No",
                                Married == 1 ~ "Yes"),
         USCitizenCat = case_when(USCitizen == 0 ~ "Not citizen",
                                  USCitizen == 1 ~ "Citizen")) %>%
prop_test(MarriedCat ~ USCitizenCat,
          order = c("Citizen", "Not citizen"), z = TRUE,
          success = "Yes")
```

```
# A tibble: 1 × 5
  statistic p_value alternative lower_ci upper_ci
      <dbl>   <dbl> <chr>          <dbl>    <dbl>
1    -0.380   0.704 two.sided     -0.101   0.0682
```

Difference in Means

Let's estimate the average hours worked per week between married and unmarried US residents.

Response variable:

Explanatory variable:

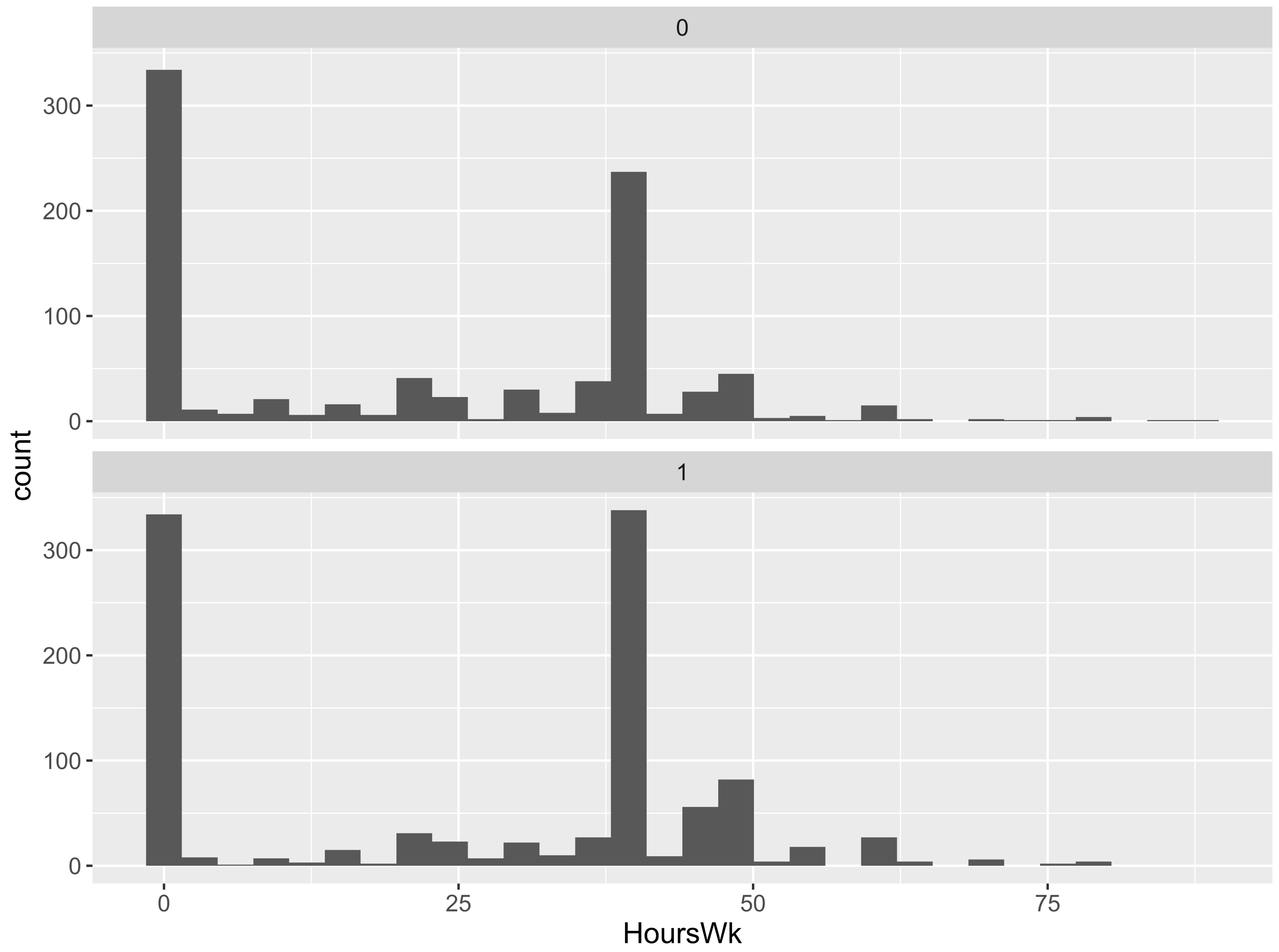
Parameter of interest:

Sample size requirement for theory-based inference:

Difference in Means

Let’s estimate the average hours worked per week between married and unmarried US residents.

```
# Exploratory data analysis
ggplot(data = ACS_adults, mapping = aes(x = HoursWk)) +
  geom_histogram() +
  facet_wrap(~Married, ncol = 1)
```




```
# Sample size
ACS_adults %>%
  drop_na(HoursWk) %>%
  count(Married)
```

```
  Married    n
1       0  896
2       1 1040
```

Difference in Means

Let's estimate the average hours worked per week between married and unmarried US residents.

Which arguments for `t_test()` reflect my research question?

```
library(infer)
ACS_adults %>%
t_test(HoursWk ~ Married, order = c("1", "0"))
```

```
# A tibble: 1 × 7
  statistic  t_df    p_value alternative estimate lower_ci
upper_ci
      <dbl> <dbl>      <dbl> <chr>          <dbl>    <dbl>
<dbl>
1      4.81 1902. 0.00000160 two.sided       4.55     2.69
6.40
```

```
library(infer)
ACS_adults %>%
t_test(HoursWk ~ Married, order = c("1", "0"),
       alternative = "greater")
```

```
# A tibble: 1 × 7
  statistic  t_df     p_value alternative estimate
lower_ci upper_ci
      <dbl> <dbl>       <dbl> <chr>          <dbl>
<dbl>    <dbl>
1      4.81 1902. 0.000000800 greater         4.55
2.99      Inf
```

Correlation

We want to determine if age and hours worked per week have a positive linear relationship.

Response variable:

Explanatory variable:

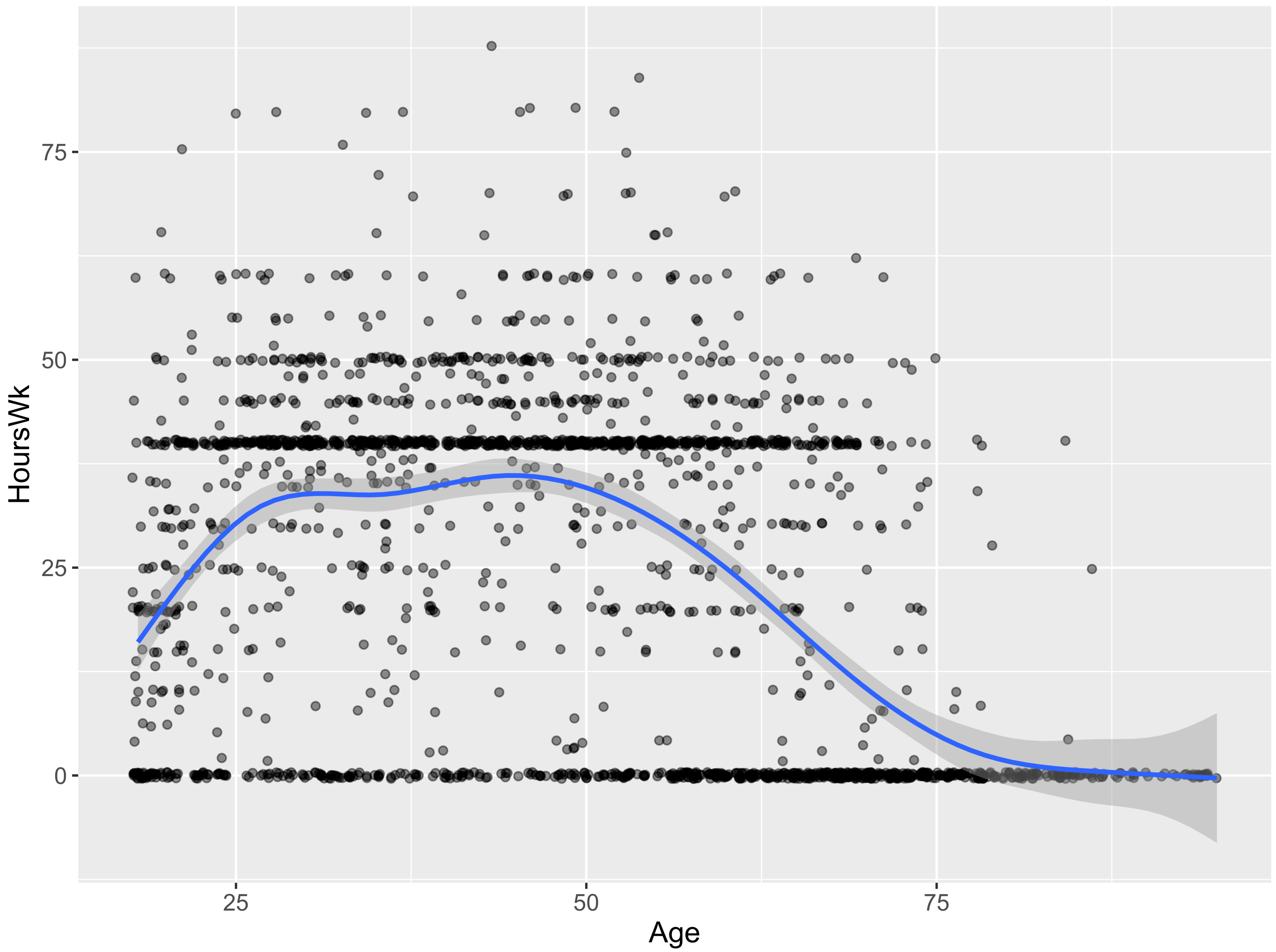
Parameter of interest:

Sample size requirement for theory-based inference:

Correlation

We want to determine if age and hours worked per week have a positive linear relationship.

```
# Exploratory data analysis
ggplot(data = ACS_adults,
       mapping = aes(x = Age,
                     y   = HoursWk)) +
  geom_jitter(alpha = 0.5) +
  geom_smooth()
```



Correlation

We want to determine if age and hours worked per week have a positive linear relationship.

```r
cor.test(~ HoursWk + Age, data = ACS_adults, alternative = "greater")
```

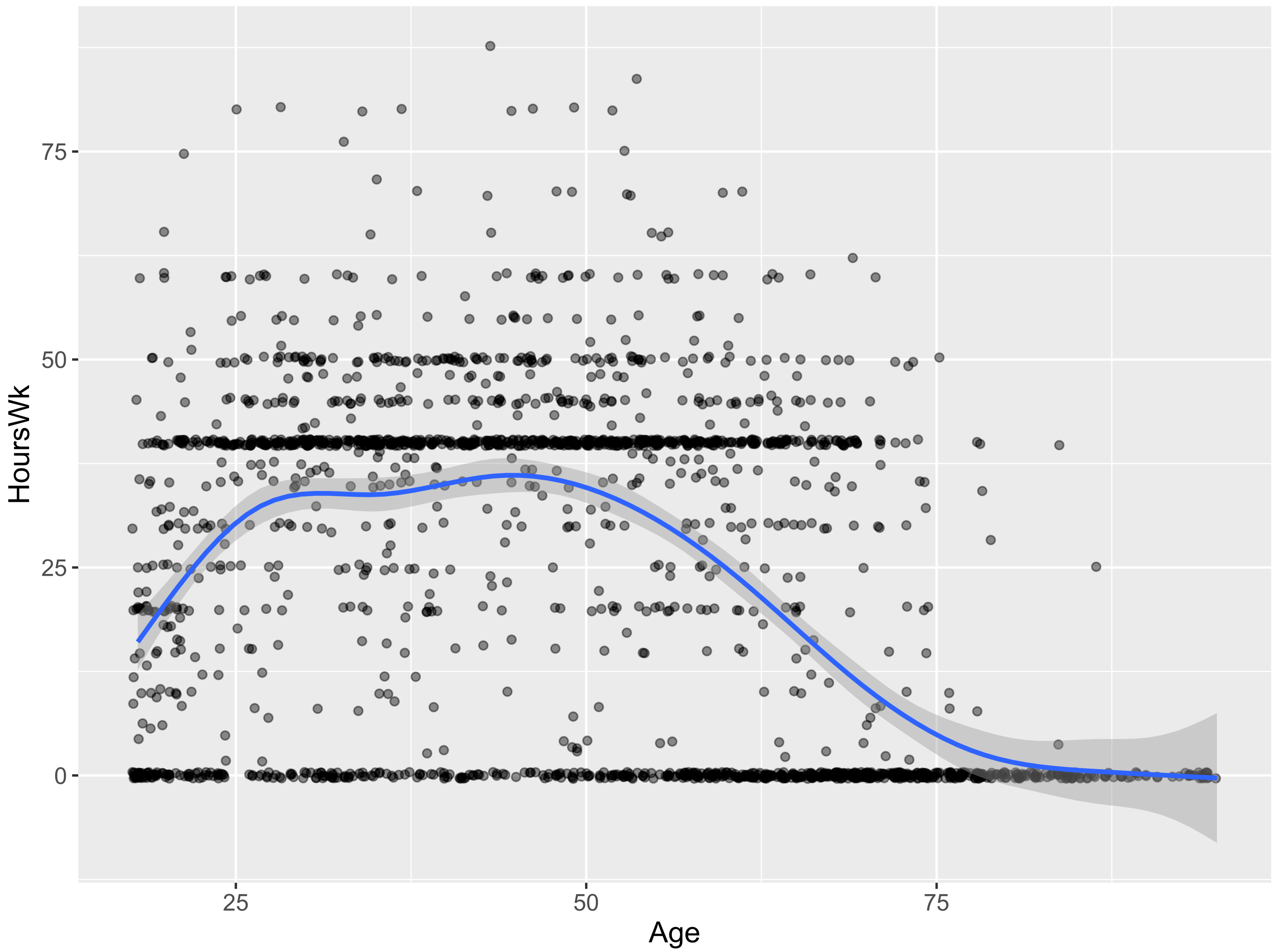
```
	Pearson's product-moment correlation

data:  HoursWk and Age
t = -17.007, df = 1934, p-value = 1
alternative hypothesis: true correlation is greater than 0
95 percent confidence interval:
 -0.3927809  1.0000000
sample estimates:
      cor 
-0.360684 
```

Correlation

We want to determine if age and hours worked per week have a positive linear relationship.

```r
# Exploratory data analysis
ggplot(data = ACS_adults,
       mapping = aes(x = Age,
                     y   = HoursWk)) +
  geom_jitter(alpha = 0.5) +
  geom_smooth()
```



Have Learned Two Routes to Statistical Inference

Which is **better**?

Is Simulation-Based Inference or Theory-Based Inference better?

Depends on how you define better.

- If better = Leads to better understanding: → Research tends to show students have a better understanding of p-values and confidence from learning simulation-based methods.
- If better = More flexible/robust to assumptions: → The simulation-based methods tend to be more flexible but that generally requires learning extensions beyond what we've seen in DATA 100.
- If better = More commonly used: → Definitely the theory-based methods but the simulation-based methods are becoming more common.
- Good to be comfortable with both as you will see both used!

What about inference for linear regression models?